SERIE of PS N°03 (PROLOG & Managing Time & Tasks)

Exercise 1: Learning to define recursive functions in **Prolog**.

Statement: We want to implement the function **Power**: $X^N = X \times X^{N-1}$

Exercise 2: Prolog Exercise: List Manipulation

Objective: Master: recursion on lists, head/tail manipulation, and standard predicates

Statement: Consider a list of integers.

1. Write a predicate `length(L, N)` that calculates the length of a list.
2. Write a predicate `sum_list(L, S)` that calculates the sum of the elements.
3. Write a predicate `max_list(L, Max)` that returns the maximum.
4. Write a predicate `member(X, L)` that checks if X belongs to the list.
5. Write a predicate `concat(L1, L2, L3)` that concatenates two lists.

Testing: max([1, 2, 15], Max), member(15, [1, 2, 15]), sum([1, 2, 15], S), member([14, 5], [1, 2, 15], L)

Exercise 3: Managing an Intelligent Schedule

Objective: Model a simple scheduling system (inspired by an AI assistant) using Prolog.

Statement: We want to represent the following tasks:

Task	Duration	Priority
article	4 h	High
course	3 h	High
meeting	2 h	Medium
correction	5 h	Medium
sports	3 h	Low

Questions:

1. Represent the **facts** in **Prolog**
2. Write a **rule**: **important(X)** if the priority is high
3. Write a **rule**: **long(X)** if the duration > 3
4. Write a **rule**: **priority(X)** if the task is important **AND** long
5. Write a query to display the priority tasks

Exercise 4: Time Optimization with "Reclaim.ai"

Objective: Understand how an AI tool like **Reclaim.ai** can optimize time management and productivity.

Statement: A professor uses Reclaim.ai to organize their week.
They have the following tasks:

Task	Duration	Priority	Deadline
Writing article	4 h	High	Friday
Preparing lecture	3 h	High	Monday
Lab meeting	2 h	Medium	Wednesday
Grading papers	5 h	Medium	Sunday
Sports	3 h	Low	Flexible

Constraints:

• Work between 9am and 5pm
• **Lunch break:** 12pm–1pm
• **Fixed lab meeting:** Wednesday 10am–12pm
• The system must balance work and well-being

Questions:

1. **Understanding:** Explain the role of **Reclaim.ai** in time management.
2. **Application:** Propose an optimized weekly schedule.
3. **Analysis:** What advantages does AI bring to this tool?
4. **Critique:** Give two limitations of using **Reclaim.ai**.

Z. Bouzidi